Influence of financial education on financial risk appetite across a variety of factors

By: Arnav Das

Purpose statement

The purpose of this study is to examine whether financial education influences regional differences in financial risk appetite across U.S. states through factors such as political affiliation, population density, and income.

Background Research

01 Financial Education

- Greater access to financial education is linked to higher risk tolerance (Chen et al., 2024)
- Builds confidence in handling financial uncertainty
- Effects are stronger when combined with higher income

02 Political Ideology

- Political ideology influences risk preferences (Jost et al., 2003)
- Conservatives tend to prefer stability, while liberals are more open to change
- Investment behavior can vary depending on political context (Kaustia & Torstila, 2011; Grable et al., 2022)

03 Population Density

- Urban residents tend to be more willing to take risks (Shi & Yan, 2017)
- Cities provide more exposure to financial markets and opportunities
- Rural populations tend to show more cautious financial behavior

04 Income

- Higher income is associated with more investment-based risk (Xie et al., 2022)
- Lower income groups tend to express risk through spending
- Wealthier individuals are more likely to invest in high-risk opportunities (Honjo et al., 2022)

Procedures

1

Data Sources

- **NFCS (2009–2021):** J2 risk tolerance scores (1–10)
- **MIT Election Lab:** Republican vote share by state
- **U.S. Census:** population density and median income
- **CEE:** financial education mandate status

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Preparation

- Grouped NFCS data by state and year (2009, 2012, 2015, 2021)
- Removed incomplete responses
- Calculate average J2 score per state
- Find median income, population density, and %vote share per state

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Variables

- **Dependent:** financial risk tolerance (J2)
- **Independent:** income, population density, vote share
- Data aligned across matching years

Mandate States: Utah, Missouri, Tennessee, Alabama, Virginia, and North Carolina

Non-Mandate States: Texas, Florida, Pennsylvania, Massachusetts, California, and New York

Procedures

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Statistical Analysis

- Used Pearson correlation coefficient (r)
- Measured strength and direction of relationships between risk tolerance and each variable
- Calculated separately for each survey year to track changes over time

State	Med Income	Pop Density	% of R Vote
Alabama	-0.29	-0.1	-0.12
California	-0.8	-0.69	0.67
Florida	-0.83	-0.85	-0.77
Massachusetts	-0.83	-0.96	0.47
Missouri	-0.07	0.12	0.34
New York	-0.52	-0.65	0.48
North Carolina	-0.63	-0.78	-0.19
Pennsylvania	-0.15	-0.08	0.24
Tennessee	0.76	0.74	0.58
Texas	0.56	0.69	0.2
Utah	0.27	0.41	0.24
Virginia	0.7	0.84	-0.97

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Regression Model

- Model included:
  - Log of population density
  - Log of median income
  - Republican vote share
- Estimated predicted risk tolerance while holding variables constant

$$RiskTolerance = \beta_0 + \beta_1 log(PopDensity) + \beta_2 log(MedianIncome) + \beta_3 RepShare + \epsilon$$

Graphs

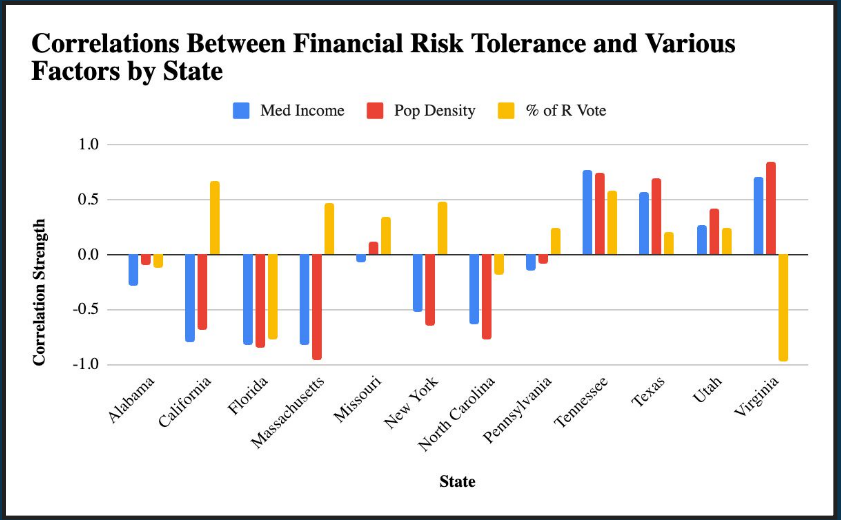
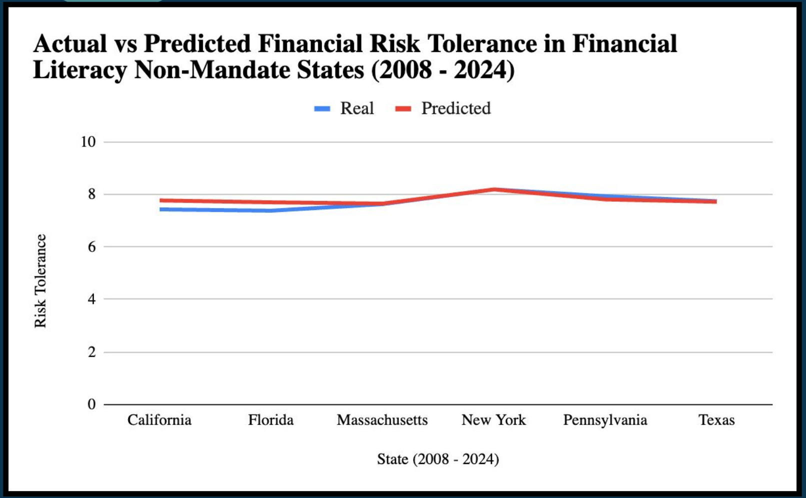


Figure 1: Pearson correlations between risk tolerance and income, population density, and vote share. Some states show strong negative relationships (CA, FL, MA, NC), while others show positive trends (TN, TX, UT, VA), indicating demographic effects vary by state

Graphs



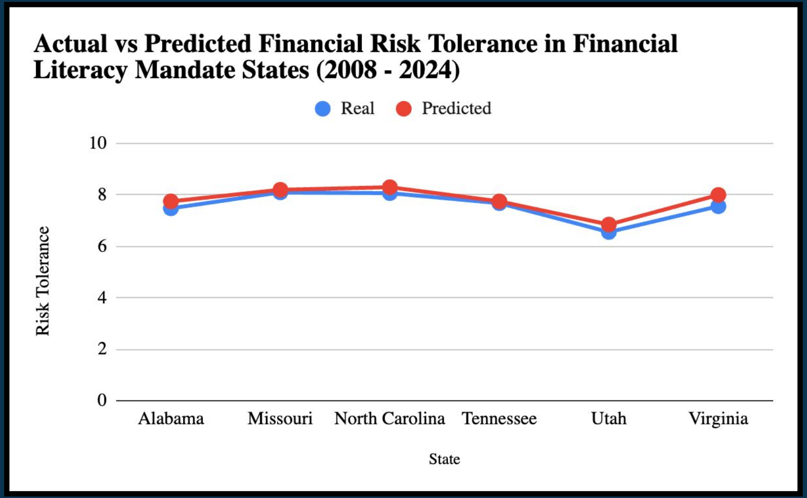


Figure 2: Comparison of actual and predicted risk tolerance in mandate states. Predicted values closely match observed averages, with small residuals (-0.07 to -0.44), indicating strong model fit.

Figure 3: Comparison of actual and predicted risk tolerance in non-mandate states. Values closely align, with small residuals (-0.34 to +0.12), though some states show slightly lower observed risk than predicted.

Data Analysis

Regression and correlation were used to examine financial risk tolerance while controlling for income, population density, and political affiliation

Across 12 states, the regression model predicted risk tolerance with **small residual error**

- Mandate states: -0.44 to -0.07
- Non-mandate states: -0.34 to +0.12
- Indicates the model explains much of the variation

	Alabama	Missouri	North Carolina	Tennessee	Utah	Virginia
Residual	-0.27	-0.1	-0.23	-0.07	-0.29	-0.44
	California	**Florida**	**Massachusetts**	**New York**	**Pennsylvania**	**Texas**
Residual	-0.34	-0.32	-0.02	0	0.12	0.02

Correlation (Demographics):

- Income and population density often show **negative relationships**
  - CA, FL, MA, NC (r ≈ -0.63 to -0.96)
- Some mandate states show **positive relationships**
  - TN (income 0.76, density 0.74)
  - VA (income 0.70, density 0.84)
- Suggests demographics alone do not consistently predict risk tolerance

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Data Analysis

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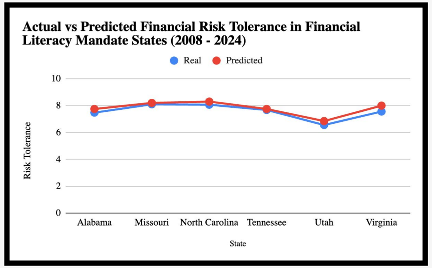
Correlation (Political):

- Republican vote share shows more consistent relationships
- Positive: TN (0.58), MO (0.34), NY (0.48)
- Negative: FL (-0.77), VA (-0.97)
- Suggests political ideology may influence financial risk behavior

Mandate Comparison:

- Mandate states show **slightly higher observed risk tolerance than predicted**
- Suggests a possible association with financial education mandates



Conclusion / Limitation:

- Mandates may be linked to higher risk tolerance
- Results are **observational** and may reflect other state-level factors or timing differences

Data Gaps

- **Sample & Data Scope**
  - Only 12 states analyzed, which limits generalizability
  - State-level averages may hide individual differences within states
- **Measurement of Risk**
  - Based on one NFCS variable (J2, 1–10 scale)
  - May not reflect actual financial behavior (reported vs real decisions)
  - Minor timing differences across datasets may affect consistency
- **Omitted Variables**
  - Model includes income, population density, and political affiliation
  - Does not include factors like education level, age, race, employment, or financial access
  - These variables may influence both financial education exposure and risk tolerance
- **Mandate Classification**
  - States grouped as mandate vs non-mandate
  - Does not account for differences in policy quality or implementation
  - Variation in course depth and effectiveness is not captured

Moving Forward

- **Expand Data Scope**
  - Include all 50 states and more years of data
  - Helps identify long-term trends and improve reliability
  - Clarifies whether observed effects are meaningful
- **Individual-Level Analysis**
  - Use surveys to measure personal financial behavior
  - Examine differences by income, political identity, and location
  - Provides more direct evidence than state-level averages
- **Measure Real Behavior**
  - Analyze outcomes like investing, saving, and entrepreneurship
  - Include indicators such as stock participation, retirement savings, and financial advisor use
  - Offers stronger evidence than self-reported risk tolerance
- **Overall Insight**
  - Financial education may increase risk tolerance slightly
  - Political, economic, and regional factors also play major roles
  - Improving outcomes requires both better access and higher quality financial education

Works Cited

Abdellaoui, M., Bleichrodt, H., & Paraschiv, C. (2007). Loss aversion under prospect theory: A parameter-free measurement. *Management Science, 53*(10), 1659–1674. https://doi.org/10.1287/mnsc.1070.0711

Gächter, S., Johnson, E. J., & Herrmann, A. (2007). Individual-level loss aversion in riskless and risky choices. *CeDEx Discussion Paper Series No. 2007-02.* https://doi.org/10.2139/ssrn.957507

Grable, J. E., Warmath, D., & Kwak, S. (2022). Financial risk tolerance before and after the 2020 U.S. presidential election. *Journal of Behavioral and Experimental Finance, 36,* 100723. https://doi.org/10.1016/j.jbef.2022.100723

Jost, J. T., Glaser, J., Kruglanski, A. W., & Sulloway, F. J. (2003). Political conservatism as motivated social cognition. *Psychological Bulletin, 129*(3), 339–375. https://doi.org/10.1037/0033-2909.129.3.339

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica, 47*(2), 263–291. https://doi.org/10.2307/1914185

Tversky, A., & Kahneman, D. (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty, 5*(4), 297–323. https://doi.org/10.1007/BF00122574

Kaustia, M., & Torstila, S. (2011). Stock market aversion? Political preferences and stock market participation. *Journal of Financial Economics, 100*(1), 98–112. https://doi.org/10.1016/j.jfineco.2010.10.017

Shi, X., & Yan, Z. (2017). *Urbanization and risk preference in China: A decomposition of self-selection and assimilation effects. China Economic Review, 45,* 231–246. https://doi.org/10.1016/j.chieco.2017.04.009

Xie, X., Tong, Z., & Xu, S. (2022). *Risk attitude and household consumption: Evidence from China Household Finance Survey data. Frontiers in Psychology, 13,* 1050371. https://doi.org/10.3389/fpsyg.2022.1050371

FINRA Investor Education Foundation. (2009–2021). *National Financial Capability Study (NFCS) state-by-state survey data.* Retrieved from https://www.usfinancialcapability.org/downloads.php

MIT Election Data and Science Lab. (2023). *U.S. president 1976–2020 and 2024 general election results.* Harvard Dataverse. Retrieved from https://dataverse.harvard.edu/dataverse/medsl

Grable, J. E., Lytton, R. H., & O'Neill, B. (2022). *Exploring the determinants of financial risk tolerance. Journal of Financial Counseling and Planning, 33*(1), 5–20. https://doi.org/10.1891/JFCP-21-0003